

QTR5

APOLLO 13	
LM DATA CARD BOOK	
PART NO	S/N
SKB 32100082-387	

FINAL

CHANGE
MARCH 30, 1970

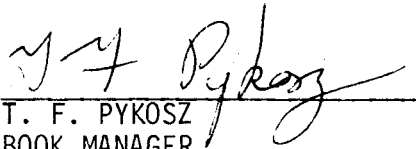
APOLLO 13

FINAL

LM DATA CARD BOOK

MARCH 30, 1970 CHANGE

PREPARED BY:


T. F. PYKOSZ
BOOK MANAGER
TRW TASK 81

MARCH 30, 1970

APPROVED BY: SIGNATURE AUTHORIZED BY TELECOM; MARCH 30, 1970
J. W. O'NEILL
CHIEF
FLIGHT PLANNING BRANCH

It is requested that any organization having comments, questions, or suggestions concerning this document contact T. F. Pykosz, Building 4, Room 278A, Telephone Number HU-3-4015.

This document is under the configuration control of the Crew Procedures Control Board (CPCB). All proposed changes should be submitted to the Apollo Flight Data File Manager, Mr. T. W. Holloway, CF34, Room 230, telephone HU3-4271.

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APOLLO FLIGHT DATA FILE

LM DATA CARD BOOK

BASIC DATE 3/17/70

CHANGE 3/30/70

LIST OF EFFECTIVE PAGES

*Indicates Current Change

PAGE NUMBER

ISSUE

1	.3/24/70
2 thru 7.	.3/17/70
8	.3/24/70
*9 thru 11	.3/30/70
12 thru 14.	.3/17/70

This LOEP reflects changes incorporated
as a result of approved changes:

027, ΔR VALUES

ACTIVATION
GYRO DRIFT COMP.[illegible][illegible]

UNDOCK/SEP GET

$$\frac{(\dots)}{(99)} \vdots \frac{(\dots)}{16} \vdots \frac{(\dots)}{27}$$

GYRO DRIFT COMPENSATION

ΔT COMPUTATION		
N93 PRO TIME	_____	
PREVIOUS TORQ	_____	
TIME	_____	
ΔT	_____	
BIAS SHIFT		
N93	BIAS SHIFT	
X	= _____ X DEG/HR	
ΔT	_____	
Y	= _____ Y DEG/HR	
ΔT	_____	
Z	= _____ Z DEG/HR	
ΔT	_____	
GYRO DRIFT		
X	Y	Z
0.0150	0.0030	0.0060
NBD (OLD)		
BIAS SHIFT		
NBD (NEW)		
OCTAL		
PROCEDURES		
V25N01E	V21N01E	
1460E	1460E	
NBDX XXXXXE	NBDX XXXXXE	E
NBDY XXXXXE	NBDY XXXXXE	E
NBDZ XXXXXE	NBDZ XXXXXE	E

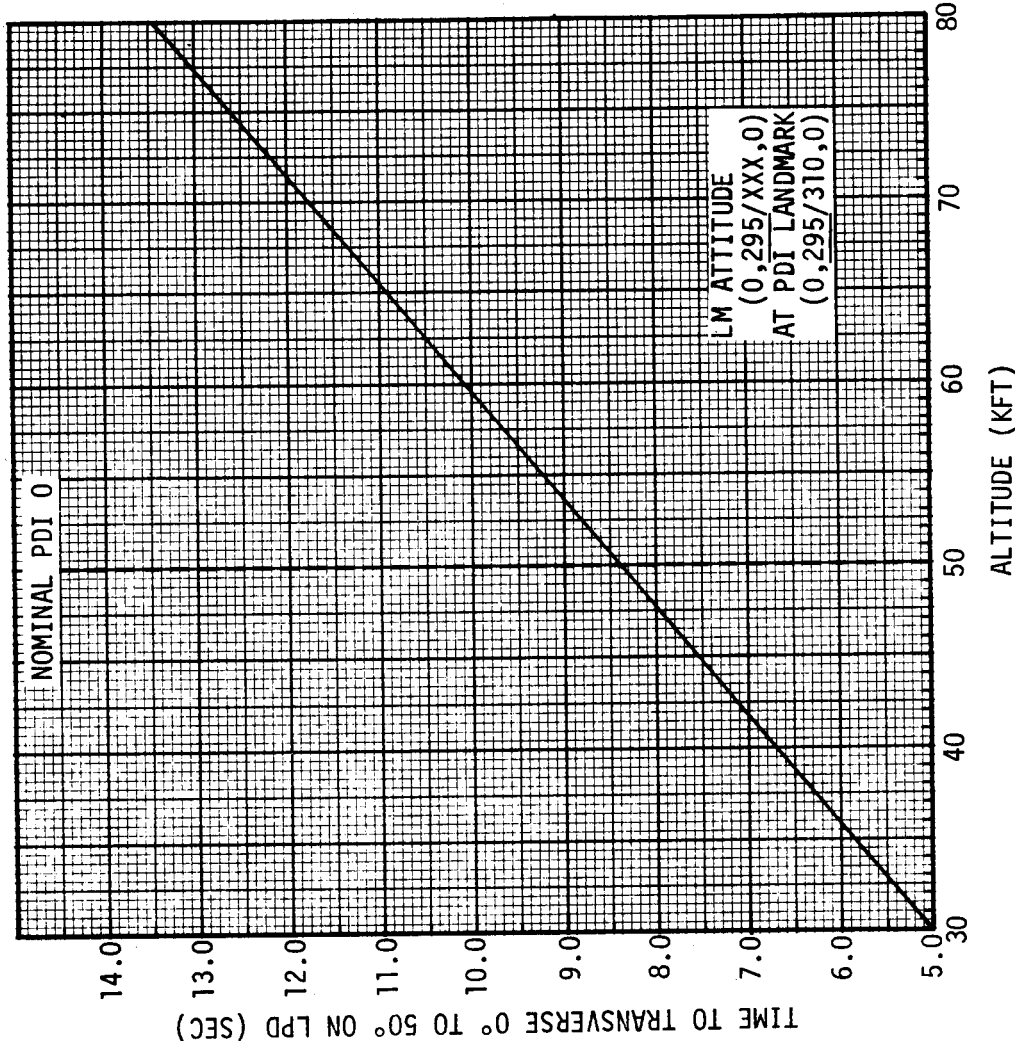
DEC/HR	OCTAL		DEC/HR	OCTAL	
	+	-		+	-
000	00000	00000	.765	14532	63245
.015	00177	77600	.780	14731	63046
.030	00377	77400	.795	15131	62646
.045	00576	77201	.810	15330	62447
.060	00775	77002	.825	15527	62250
.075	01174	76603	.840	15726	62051
.090	01374	76403	.855	16126	61651
.105	01573	76204	.870	16325	61452
.120	01772	76005	.885	16524	61253
.135	02171	75606	.900	16723	61054
.150	02371	75406	.915	17123	60654
.165	02570	75207	.930	17322	60455
.180	02767	75010	.945	17521	60256
.195	03166	74611	.960	17721	60056
.210	03366	74411	.975	20120	57657
.225	03565	74212	.990	20317	57460
.240	03764	74013	1.005	20516	57261
.255	04163	73614	1.020	20716	57061
.270	04363	73414	1.035	21115	56662
.285	04562	73215	1.050	21314	56463
.300	04761	73016	1.065	21513	56264
.315	05160	72617	1.080	21713	56064
.330	05360	72417	1.095	22112	55665
.345	05557	72220	1.110	22311	55466
.360	05756	72021	1.125	22510	55267
.375	06155	71622	1.140	22710	55067
.390	06355	71422	1.115	23107	54670
.405	06554	71223	1.170	23306	54471
.420	06753	71024	1.185	23505	54272
.435	07152	70625	1.200	23705	54072
.450	07352	70425	1.215	24104	53673
.465	07551	70226	1.230	24303	53474
.480	07750	70027	1.245	24502	53275
.495	10150	67627	1.260	24702	53075
.510	10347	67430	1.275	25101	52676
.525	10546	67231	1.290	25300	52477
.540	10745	67032	1.305	25477	52300
.555	11145	66632	1.320	25677	52100
.570	11344	66433	1.335	26076	51701
.585	11543	66234	1.350	26275	51502
.600	11742	66035	1.365	26474	51303
.615	12142	65635	1.380	26674	51103
.630	12341	65436	1.395	27073	50704
.645	12540	65237	1.410	27272	50505
.660	12737	65040	1.425	27472	50305
.675	13137	64640	1.440	27671	50106
.690	13336	64441	1.455	30070	47707
.705	13535	64242	1.470	30267	47510
.720	13734	64043	1.485	30467	47310
.735	14134	63643	1.500	30666	47111
.750	14333	63444			

DATE MARCH 17, 1970

LM DATA CARD BOOK

ALTITUDE CHECK/PDI RULES CARD

ALTITUDE DETERMINATION CHART



PREPARED BY FPRB/OPS
MISSION APOLLO 13, FEBRUARY 3, 1970

PDI RULES

1. NO ULLAGE-NO GO FOR PDI.
2. NO IGNITION DELAY 2 SEC THEN START PB-PUSH; THEN SET DES ENG OVRD-ON AT 5 SEC.
3. T/W >1.6 AND DSKY CHANGES >18 fps/2 SEC.
4. ATT/RATE <5°/SEC.
5. ΔH WITHIN LIMITS >10 SEC AND NOT OUT OF LIMITS >60 SEC.
6. DATA GOOD AT >10,000 FT.
7. IF NO THROTTLE DOWN BY P64 +15 SEC-ABORT.
8. BINGO FUEL 1 MIN 34 SEC AFTER LOW LEVEL OR WHEN FUEL QTY <2% UNLESS LANDING IMMINENT.

NOTE: FOR FLASHING LR ALT OR VEL LIGHTS THAT ARE PRECEDED BY A STEADY LR LIGHT, CYCLE THE RADAR TEST SWITCH.

CSM CIRC BURN

P76

+	0	0				+	0	0	1	0	0	HRS	N33
+	0	0	0			+	0	0	0	3	5	MIN	TIG
+	0					+	0	0	5	0	0	SEC	
						+	0	0	6	9	0	ΔVX	N84
						+	0	0	0	0	0	ΔVY	
						+	0	0	1	4	0	ΔVZ	

CSM HA/HP

CMC		LGC	V82
/		/	OPT 2

PDI 0/NO PDI + 12 CARD

PAGE 4

APOLLO 13 FLIGHT DATA FILE

PDI 0 ABORT PAD

+	0	0				+	0	0	1	0	1	HRS	N33
+	0	0	0			+	0	0	0	0	3	MIN	TIG
+	0					+	0	5	3	4	0	SEC	A
						+	0	1	0	0	0	AVX	N81
						+	0	0	0	0	0	AVY	LV
						+	0	0	0	1	4	AVZ	B
+						+	0	1	3	6	3	HA	N42
						+	0	0	0	8	4	HP	
+						+	0	1	0	0	0	AVR	
X	X	X				X	X	X	0	3	6	BT	
X	X	X				X	X	X	1	8	1	R	FDAI
X	X	X				X	X	X	0	0	3	P	INER
						+	0	1	0	0	0	AVX	N86
						+	0	0	0	0	0	AVY	AGS
						+	0	0	0	3	4	AVZ	
+	0	0				+	0	0	1	0	2	HRS	N11
+	0	0	0			+	0	0	0	0	3	MIN	CSI
+	0					+	0	2	8	6	0	SEC	C
+	0	0				+	0	0	1	0	4	HRS	N37
+	0	0	0			+	0	0	0	0	2	MIN	TPI
+	0					+	0	1	1	3	0	SEC	D

RESIDUALS

PGNS				AGS			
	AVX	N85		AVX	500		
	AVY			AVY	501		
	AVZ			AVZ	502		

BURN TIME IF > 1 SEC : _____
DATE MARCH 17, 1970

NO PDI + 12 ABORT PAD

+	0	0				+	0	0	1	0	3	HRS	N33
+	0	0	0			+	0	0	0	4	2	MIN	TIG
+	0					+	0	0	6	5	0	SEC	E
						+	0	1	1	0	2	AVX	N81
						+	0	0	0	0	0	AVY	LV
						-	0	0	4	8	6	AVZ	F
+						+	0	1	4	3	9	HA	N42
						+	0	0	0	8	9	HP	
+						+	0	1	2	0	4	AVR	
X	X	X				X	X	X	0	3	8	BT	
X	X	X				X	X	X	1	8	2	R	FDAI
X	X	X				X	X	X	0	0	2	P	INER
						+	0	1	1	1	2	AVX	N86
						+	0	0	0	0	0	AVY	AGS
						-	0	0	4	6	1	AVZ	
+	0	0				+	0	0	1	0	6	HRS	N11
+	0	0	0			+	0	0	0	0	3	MIN	CSI
+	0					+	0	2	1	4	0	SEC	G
+	0	0				+	0	0	1	0	8	HRS	N37
+	0	0	0			+	0	0	0	0	1	MIN	TPI
+	0					+	0	4	4	5	0	SEC	H

RESIDUALS

PGNS				AGS			
	AVX	N85		AVX	500		
	AVY			AVY	501		
	AVZ			AVZ	502		

BURN TIME IF > 1 SEC : _____
LM DATA CARD BOOK

PDI 1 PAD

($1 < \text{PDI} \leq 5.40$) ABORT PAD EARLY

$$\begin{array}{r}
\text{LOG INSERTION GET=} \\
+ \quad \frac{\cdot}{\cdot} \quad \frac{\cdot}{\cdot} \quad \frac{\cdot}{\cdot} \\
\text{BOOST GET=} \quad \frac{1}{1} \quad \frac{0}{0} \quad \frac{0}{0} \quad \frac{0}{0} \\
+ \quad \frac{\cdot}{\cdot} \quad \frac{\cdot}{\cdot} \quad \frac{\cdot}{\cdot} \\
\text{HAM GET=} \quad \frac{1}{1} \quad \frac{0}{0} \quad \frac{0}{0} \quad \frac{0}{0} \\
+ \quad \frac{\cdot}{\cdot} \quad \frac{\cdot}{\cdot} \quad \frac{\cdot}{\cdot} \\
\text{CSI GET=} \quad \frac{1}{1} \quad \frac{0}{0} \quad \frac{0}{0} \quad \frac{0}{0}
\end{array}$$

TT1-1 (5:40 <PDI 1 ≤ 15) ABORT PAD LATE

$$\begin{array}{r} \text{LOG INSERTION GET=} \\ + \\ \text{CSI TIG=} \end{array}$$
[illegible][illegible]

T2-1 AT PDI +

20:45

N69							
							ΔDN RNG
							ΔX RNG
							ARLS

THROTTLE DOWN

N43					
					LAT (+N)
					LONG (+E)
					ALT

PDI 2 ABORT CARD

PDI 1/PDI 1 ABORT
PDI 2/PDI 2 ABORT

PDI 2 PAD													
+	0	0				+	0	0	1	0	5	HRS	N33
+	0	0	0			+	0	0	0	2	4	MIN	PDI
+	0					+	0	1	2	7	0	SEC	
X	X					X	X					TGO	N61
												CROSSRANGE	
X	X	X				X	X	X	X			R	FDAI
X	X	X				X	X	X	X			P	AT TIG
X	X	X				X	X	X				Y	
												DEDA 231 IF RQD	
(1 <PDI2 ≤8:30) ABORT PAD EARLY													
LOG INSERTION GET= _____ : _____ : _____													
+ _____ 1 _____ 0 _____ 0 _____													
BOOST GET= _____ : _____ : _____													
+ _____ 1 _____ 0 _____ 0 _____													
HAM GET= _____ : _____ : _____													
+ _____ 1 _____ 0 _____ 0 _____													
CSI GET= _____ : _____ : _____													
+ 0 0 _____ + 0 0 1 1 0 _____ HRS N37													
+ 0 0 0 _____ + 0 0 0 0 1 7 _____ MIN TPI													
+ 0 _____ + 0 0 0 5 7 0 _____ SEC													
T1-2 (8:30 ≤PDI 2 ≤ 15) ABORT PAD LATE													
LOG INSERTION GET= _____ : _____ : _____													
+ _____ 5 _____ 5 _____ 0 _____													
CSI TIG= _____ : _____ : _____													
+ 0 0 _____ + 0 0 0 1 0 8 _____ HRS N37													
+ 0 0 0 _____ + 0 0 0 0 1 8 _____ MIN TPI													
+ 0 _____ + 0 0 4 7 6 0 _____ SEC													

T2-2 (PDI 2+18:56) ABORT PAD											
LOG INSERTION GET= _____ : _____ : _____											
+ _____ 5 _____ 0 _____ 0 _____											
CSI GET= _____ : _____ : _____											
+ 0 0 _____ + 0 0 0 1 0 5 _____ HRS N33											
+ 0 0 0 _____ + 0 0 0 0 4 3 _____ MIN TIG											
+ 0 _____ + 0 0 0 9 3 0 _____ SEC											
+ 0 0 _____ + 0 0 0 1 0 8 _____ HRS N37											
+ 0 0 0 _____ + 0 0 0 0 1 8 _____ MIN TPI											
+ 0 _____ + 0 0 4 8 0 0 _____ SEC											
T2-2 AT PDI + _____ 18:56											
N69											
ADN RNG											
ΔX RNG											
ΔRLS											
THROTTLE DOWN _____ : _____											
N43											
LAT (+N)											
LONG (+E)											
ALT											

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LM DATA CARD BOOK

DATE MARCH 17, 1970

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APOLLO 13

FLIGHT DATA FILE

PAGE 7

FIRST REV ACTIVITY		G&N LUNAR SURFACE CARD		LAUNCH PREP	
N20	OG	047	053	P57, A/T 3 LANDING SITE	377
	IG	544	+5:02	N04	544
	MG	545		STAR	545
		546			546
P57, A/T 1, REFSMMAT		232	465	CURS	514
N04	TILT			V32	515
N05				CURS	516
N93	X Y Z			V32	
				CURS	
P57 A/T 2, REFSMMAT				N05	047
STAR1					053
CURS					
V32					
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ABORT/ASCENT CARD

ASCENT RULES

UNDERBURN			PGNS		AGS	
ΔV FPS	TIME SEC					
<400	20		NULL RESIDUALS		AUTO, A/H 15fps	
>400	20		A/H BURN Hp		AUTO, A/H 15fps	

INSERTION

AGS AND PGNS RESIDUALS AGREE WITHIN 10FPS,
TRIM TO LESS THAN 2FPS (AGS X-AXIS ONLY)
AND STANDBY FOR TWEAK.
TWEAK AT INSERTION PLUS 4 MINUTES
(10° OHW OR 250° FDAI)

FOR NO VOICE

PGNS, AGS DIFFER <10FPS, TRIM ACTIVE SYSTEM
PGNS, AGS DIFFER >10FPS, TRIM SYSTEM WHICH
AGREES WITH RR
ATT/RATE ERROR >10°/SEC

T3 (1 REV) ABORT PAD

LOG INSERTION GET=

+

CSI TIG=

+

TPI TIG=

+	0	0			+	0	0	1	0	5		HRS	N33
+	0	0	0		+	0	0	0	4	1		MIN	TIG
+	0				+	0	3	7	5	0		SEC	

P22 ACQUISITION TIME

____:____:____
____:____:____

DATE MARCH 24, 1970

LM ASCENT PAD

+	0	0	1	3	7	+	0	0	1	3	7	HRS	N33
+	0	0	0	1	2	+	0	0	0	0	9	MIN	TIG
+	0	5	3	0	0	+	0	1	5	8	1	SEC	
+	5	5	4	0	2	+	5	5	3	3	8	V (HOR)	
+	0	0	3	2	4	+	0	0	3	6	3	V (VERT) N76	
+	0	0	0	0	0	+	0	0	0	0	0	*CROSSRANGE	
+	3	7	7	5	0	+	0	0	0	0	0	047	
-	7	6	2	3	3	+	4	0	0	0	0	053	
+	5	8	6	7	1	+	5	8	5	8	1	224/226	
+	5	6	9	7	8	+	5	6	9	7	8	231	
+	0	0	3	2	4	+	0	0	3	6	5	465	
+	0	0	1	3	7	+	0	0	1	3	9	HRS	N37
+	0	0	0	5	8	+	0	0	0	4	5	MIN	TPI
+	0	1	2	0	0	+	0	4	0	6	0	SEC	
+	1	0	7	7	5	+	1	0	6	8	9	LM WT	
+	3	5	6	3	9	+	3	5	6	7	1	CSM WT	

*NOTE: LOAD 8 NM CROSSRANGE IF GREATER THAN 8 NM
COMMENTS:

RESIDUALS

+					HA	V82	+					HA	315
+					HP		+					HP	403
PGNS							AGS						
						N85							
						ΔVX						ΔVX	500
						ΔVY						ΔVY	501
						ΔVZ						ΔVZ	502

+	0	0				+	0	0	1	3	8	HRS	N11
+	0	0	0			+	0	0	0	0	6	MIN	CSI
+	0					+	0	0	0	9	0	SEC	
R1(+00001), R2(+02660), R3(+13000)													N55
+	0	0				+	0	0	1	3	9	HRS	N37
+	0	0	0			+	0	0	0	4	5	MIN	TPI
+	0					+	0	4	0	6	0	SEC	
+	0					+	0	0	4	9	6	ΔVX	N81
+	0					+	0	0	0	0	0	ΔVY	LV

+						+	0	4	8	6	0	373
+						+	0	5	8	5	8	275
+	0					+	0	0	4	9	6	ΔVX
+	0					+	0	0	0	0	0	ΔVY
+	0					+	0	0	0	0	0	ΔVZ

PGZUS			
N75 CSI			
$\frac{\Delta H}{(15.0)}$	$\frac{CSI/CDH}{(58:13)}$	$\frac{CDH/TPI}{(41:27)}$	$\frac{\Delta VX}{(49.6)}$
$\frac{\cdot}{\cdot}$	$\frac{\cdot}{\cdot}$	$\frac{\cdot}{\cdot}$	$\frac{\cdot}{\cdot}$
$\frac{\cdot}{\cdot}$	$\frac{\cdot}{\cdot}$	$\frac{\cdot}{\cdot}$	$\frac{\cdot}{\cdot}$
$\frac{\cdot}{\cdot}$	$\frac{\cdot}{\cdot}$	$\frac{\cdot}{\cdot}$	$\frac{\cdot}{\cdot}$

AGS	$\frac{402}{\Delta H}$	$\frac{372}{\text{CSI/CDH}}$	$\frac{267/450}{\Delta V G \text{ (CSI)}}$	CSM SOLUTION
	_____	_____	_____	ΔVX CSI
	_____	_____	_____	•
	_____	_____	_____	•
	_____	_____	_____	•

BURN RULES

CRITERIA IS $\dot{X}=3\text{fps}$
PRIORITY OF SOLUTIONS-PGNS, AGS, CMC, CHARTS.

A. RR AGREES WITH VHF WHERE
$$\Delta R = \frac{R}{100} + 0.5\text{NM}, \Delta R \text{ IS ALWAYS } \geq 1\text{NM}$$

IF TWO OF THREE SOLUTIONS AGREE,
BURN THE PRIORITY SOLUTION.

B. RR DOES NOT AGREE WITH VHF.
MSFN ISOLATES FAILED SYSTEM.

C. V90 5fps-NO BURN.

MAX N49 (2.0,12.0)			RESIDUALS		
			PGNS		AGS
PGNS	AGS			ΔVX	ΔVX 500
		HA		ΔVY	ΔVY 501
		HP		ΔVZ	ΔVZ 502

N81 CSI	N82 CDH	P GZCS
YD0T(1190) (+0.0)	$\frac{\Delta VX}{(+0.0)}$	$\frac{\Delta VZ}{(+0.0)}$
(-)	.	.
(-)	.	.

TION (CHANGE SIGN)		AGS	
ΔVY CSI	BIAS $\Delta VX = -1.0$	263	371
_____	_____	YDOT CSI	ΔV CDH
_____	_____	_____	_____
_____	_____	_____	_____

CDH/PLANE CHANGE CARD

+	0	0				+	0	0	1	3	9	HRS	N13
+	0	0	0			+	0	0	0	0	4	MIN	CDH
+	0					+	0	1	4	1	0	SEC	
0						+	0	0	0	0	0	ΔVX	N81
0						+	0	0	0	0	0	ΔVY	LV
0						+	0	0	0	0	0	ΔVZ	
0						+	0	0	0	0	0	PLM	FDAI
+						+	0	5	4	4	2	373	
0						+	0	0	0	0	0	ΔVX	N86
0						+	0	0	0	0	0	ΔVY	AGS
0						+	0	0	0	0	0	ΔVZ	

PLANE CHANGE P30, V90, 410+5											
TIG	CDH										
TIG	PC										
YDOT											
CSM (N90)			PGNS (N90)			AGS (270)					
(-) _____			(-) _____			(-) _____					
(-) _____			(-) _____			(-) _____					

MAX N49 (0.8,5.0)

RESIDUALS

PGNS		AGS	
ΔH	ΔT TPI/CDH	TPI SLIP	
(15.0)	(41:27)	(0:00)	
HA	HP	ΔVX	N85
		ΔVY	500
		ΔVZ	501
			502

PGNS		AGS	
ΔVX	N85	ΔVX	500
ΔVY		ΔVY	501
ΔVZ		ΔVZ	502

PGNS

PGNS

N75 CDH		N81 CDH	
ΔH	ΔT TPI/CDH	ΔVX	YDOT N90
(15.0)	(41:27)	(+0.0)	(+0.0)
		(-)	(-)
		(-)	(-)
		(-)	(-)

N75 CDH		N81 CDH	
ΔH	ΔT TPI/CDH	ΔVX	YDOT N90
(15.0)	(41:27)	(+0.0)	(+0.0)
		(-)	(-)
		(-)	(-)
		(-)	(-)

AGS

AGS

402	450	452	
ΔH	ΔVX	ΔVZ	

CSM SOLUTION (CHANGE SIGN)			
ΔVX	ΔVY	ΔVZ	

CRITERIA IS X=2fps, Z=6fps
PRIORITY OF SOLUTIONS-PGNS, AGS, CMC
CHARTS.
A. RR AGREES WITH VHF WHERE
ΔR= R +0.5NM, ΔR IS ALWAYS ≥1
100
IF TWO OF THREE SOLUTIONS
AGREE, BURN THE PRIORITY
SOLUTION.
B. RR DOES NOT AGREE WITH VHF.
MSFN ISOLATES FAILED SYSTEM.
C. V90<5fps-NO BURN.

263 270
ΔVY (CDH) ΔVY (NOM)

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TPI CARD

	+	0	0				+	0	0	1	3	9	HRS	N37
	+	0	0	0			+	0	0	0	4	5	MIN	TPI
	+	0					+	0	4	0	6	0	SEC	
R1(+00000), R2(+02660), R3(+13000)	N55													
+	0						+	0	0	2	1	9	ΔVX	N81
+	0						+	0	0	0	0	1	ΔVY	LV
-	0						-	0	0	1	1	0	ΔVZ	
+	0						+	0	3	7	7	7	R	N54
-	0						-	0	1	1	1	5	ΔVZ	TIG-5
+	0						+	0	0	2	4	5	ΔVF ⁺ /A ⁻	N59
+	0						+	0	0	0	0	1	ΔVR ⁺ /L ⁻	L0S
-	0						-	0	0	0	0	1	ΔVD ⁺ /U ⁻	
X	X						X	X	0	0	2	2	BT	
307+043.00														

BURN RULES

CRITERIA IS $\dot{x}=2\text{fps}$, $\dot{y}=5\text{fps}$, $\dot{z}=6\text{fps}$
 PRIORITY OF SOLUTIONS-PGNS, AGS, CMC, CHARTS.

A. RR AGREES WITH VHF WHERE
 $\Delta R = \frac{100}{R} + 0.5\text{NM}$, ΔR IS ALWAYS $\geq 1\text{NM}$

IF TWO OF THREE SOLUTIONS AGREE,
 BURN THE PRIORITY SOLUTION.

B. RR DOES NOT AGREE WITH VHF.
 MSFN ISOLATES FAILED SYSTEM.

RESIDUALS

MAX N49 (0.8, 5.0)

PGNS			AGS		
PGNS	AGS	HA	HP	ΔVX	ΔVY
				1	1
				3	0
				1	2

PGNCS

N37 TPI			N58 TPI			N81 TPI			N59 TPI		
TIG	(139:45:41)	21.8°	HP	ΔV	(24.7)	ΔV TPF	ΔVX	ΔVY	ΔVZ	ΔVF ⁺ /A ⁻	ΔVR ⁺ /L ⁻
137	: 55	: 12	(43.3)	44.4	104.9	46.3	(+21.9)	(+0.1)	(-11.0)	(24.5)	(+0.1)
	:	:		44.4	104.8	40.5					
	:	:									

PGNCS

AGS

N37 TPI			N58 TPI			N81 TPI			N59 TPI		
TIG	(139:45:41)	21.8°	HP	ΔV	(24.7)	ΔV TPF	ΔVX	ΔVY	ΔVZ	ΔVF ⁺ /A ⁻	ΔVR ⁺ /L ⁻
373	: 55	: 12	(43.3)	44.4	104.9	46.3	(+21.9)	(+0.1)	(-11.0)	(24.5)	(+0.1)
	:	:		44.4	104.8	40.5					
	:	:									

AGS

TPI
P76/P27

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[illegible]

AGS SV
IMPACT PAD

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